Automatic Aug-Aware Contrastive Proposal Encoding for Few-Shot Object Detection of Remote Sensing Images

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***Abstract*— In the annotation of remote sensing images (RSIs), the effectiveness of common object detection methods trained on only a few samples decreases instantly, which has prompted increasing research on the few-shot problem in remote sensing. RSIs often exhibit suboptimal performance in few-shot scenarios due to the intricate nature of scene information interference and the high degree of cosine similarity, both of which present significant challenges to their effectiveness. In this article, a two-stage detection framework based on fine-tuning is selected to deal with the common problems in the few-shot task of the remote sensing domain. Considering the excessive scale variation in instances in the remote sensing datasets, we introduce an automatically learned aug-aware search module to provide an intelligent data augmentation solution for Faster R-CNN using different optimal augmentation policies searched by the network to fit the current dataset. We introduce a contrastive RoI branch to better classify novel class proposal features that are easily confused by the base class. We named our work augmentation-aware contrastive proposal encoding (AACE) and conducted extensive experiments on two common object detection datasets in remote sensing, NWPU VHR 10 and DIOR, on which AACE achieved about 2.30% and 2.61% improvement, respectively, in the number of shots listed in the article, compared with other algorithms.**

***Index Terms*— Aug-aware search, augmentation policy, contrastive learning RoI, few-shot, proposal features, scale variation.**

I. Introduction

DEEP neural networks [23], [24] have seen much success in the field of computer vision. The application of CNNs is particularly important for detection tasks, which require both accurate classification and precise localization, which places higher demands on deep learning detectors, such as Faster R-CNN [8], you-only-look-once (YOLO) [16], and others [44], [45], [46], [55], [56], [57], [58], [59], which localize and classify targets and perform object detection tasks. However, with insufficient training samples, or when manual labeling is too expensive, few-shot learning comes into play. Few-shot detectors can only be inherited from previous models and extended to new visual concepts with a small number of novel classes, and the ability to recognize new classes poses a significant challenge in the design of such models.

A previous approach used to solve for few-shot detection was the method of meta-learners [27], [29]. During the whole process, the few-shot detection task is simulated by pairing a large number of base class samples (During pretraining, the network was fed with a large number of base class samples, and novel classes were not involved in the training.) with a small number of novel class samples. Another simple and effective two-stage fine-tuning approach (TFA) [4] pretrains a large number of base class samples, and then freezes only the parameters trained in the base class during the fine-tune stage, using a small amount of novel class data to train new box classifiers and regressors. Compared with meta-learner, TFA is simplified in terms of workflow and data preparation and does not require constructing sample pairs (episode) during the training process. However, in few-shot detection, inaccurate localization of the novel class is likely to be attributed to the involvement of the base class, and a small number of targets are likely to be classified as background. The introduction of the Faster R-CNN framework with the region proposal network (RPN) has greatly improved the problem of inaccurate localization, and object boxes returned by the model are often accurately localized [6]. However, when random selecting the remote sensing images (RSIs) in our experiments, we found that as shown in Fig. 1, in TFA, we present evidence in this work that objects are properly localized by Faster R-CNN, but still not properly classified—an important limitation, because the similarity between the various classes in the remote sensing dataset is very similar, and the classification decision boundary of the network is blurred. In particular, in few-shot learning where only a small number of samples from the novel class are included in training, it is more difficult for the network to learn enough information for classification. FSCE [6] emerged as a solution to this problem, using ideas from contrastive learning to cluster the proposal features derived from the RPN, clarifying the boundaries between classes.

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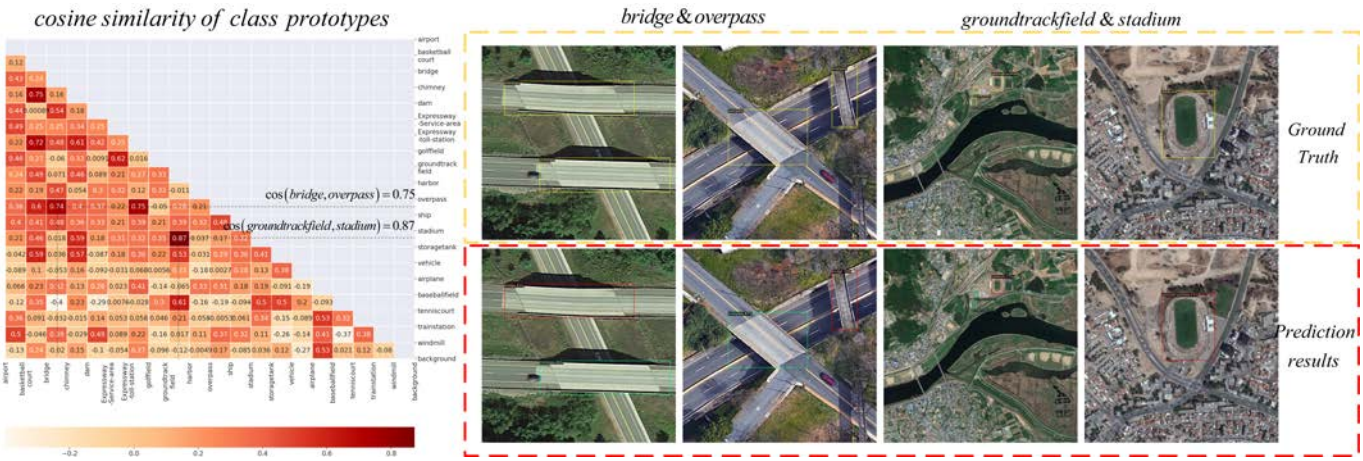


Fig. 1. (Left) Model learns cosine similarity of class prototype features guided by the loss function using about 100 images in the DIOR dataset. Similarity between classes in the remote sensing datasets is often high; we select two confusing class pairs, bridge and overpass, and ground track field and stadium, with a similarity of 0.75 and 0.87, respectively. (Right) Performance in actual detection. Yellow boxes: true labels. Red boxes: incorrect recognition. Green boxes: correct recognition. This indicates that RSI object detection suffers often from accurate localization but incorrect classification between classes due to close proximity in feature space.

It is well-known that data augmentation can provide more information to the network and make its performance more robust. For the remote sensing datasets with more complex semantic information, an effective augmentation approach can bring significant benefits. In the face of excessive individual size differences in the dataset, a feature pyramid network (FPN) structure is proposed in the object detection Faster R-CNN architecture, which is a good solution to the common scale problem in the remote sensing datasets. However, for the FPN aspect of data augmentation, it is often set manually and basically consists of simple rotation and cropping. We use an intelligent means of augmenting perception [9], uncovering suitable augmentation strategies for remote sensing data through constant search, adding them to our few-shot object detection (FSOD) task, and combining them with the Faster R-CNN FPN structure chosen for this article.

We call this automatic augmentation-aware contrastive proposal encoding (AACE)-based few-shot detection method. We first obtain the searched augmentation strategy, and with its enrichment, the network starts phase 1 pretraining against a large amount of base class data and uses only a small number of novel and base classes for fine-tuning in phase 2. The contributions of our article can be summarized as follows.

1) We use an intelligent data-enhanced search algorithm to provide rich semantic information to the two-stage fine-tune FSOD algorithm, enhancing the stability of the model.
2) In an aug-strategy based on augmented sensing, we analyze the impact of different means of augmentation on the remote sensing datasets and select a combination of augmentation policies belonging to the remote sensing domain to meet the requirements.
3) We introduce a contrastive learning RoI module to classify proposal features into positive and negative samples [6]. By adjusting the distance between samples in the feature space, a clustering effect of closer samples of the same class and further samples between different classes is achieved.
4) We applied the proposed framework to two commonly used object detection datasets, NWPU VHR 10 and DIOR, and achieved good results in the detection of few-sample novel and base classes in the few-shot task.

II. Related Work

A. Few-Shot Object Detection

1) Few-Shot Object Detection: The FSOD completes the object detection task with a small number of annotated samples. The main types of FSOD algorithms are the metric-, meta-, and fine-tune-based learning methods. RepMet [13] is a representative metric-learning method, where the backbone, embedding space, and representatives of each class are trained together, and the distance between the embedding space vector and the class representatives is used to perform object classification. In contrast, a meta-learning approach, FSRW [11], uses a redistribution module to transform a few novel class support samples into global vectors. Redmon et al. [12] added a YOLOv2 framework to the reweighting module and used support images to reweight the query features. Meta R-CNN [36] also follows this paradigm, building on Faster R-CNN, with object attention introducing additional meta-loss vectors and a redesigned prediction head. The fine-tune-based methods can often achieve better performance with a simple process. For example, TFA [4] is based on the Faster R-CNN, trains on the base class, and uses a small number of samples in the fine-tune phase to achieve detection on the novel class. FSCE [6] uses a contrastive classification head to cluster confused proposal embedding in the feature space by pulling similar sample pairs closer together, while pushing dissimilar sample pairs farther apart. As a two-stage fine-tuned few-shot algorithm, although the steps can be more complicated, it successfully introduces the idea of contrastive

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learning into the few-shot task of natural images. Inspired by FSCE, to tackle the few-shot problem in the field of remote sensing, we further explore the introduction of contrastive learning.

The above methods have all achieved good results on natural images, and for remote sensing, scholars are devising few-shot algorithms to solve difficult problems, taking into account the characteristics of the datasets.

2) FSOD for RSIs: The current remote sensing datasets often suffer from uneven sample sizes and high annotation labor costs. Much work has focused on the traditional basic classification tasks [51], [52], [53], [54], often overlooking the FSOD task in remote sensing. The main tasks of target classification are classification and localization, and in the few-shot, a small number of novel samples involved in training often cause overfitting of the classification effect to a large amount of base class data. PAMS-Det [21] replaces convolution in the network with involution and uses a path-aggregation module to modify the FPN structure. FSODM [32] uses a reassignment weight module and a reassignment weight vector extracted from the support image for learning. P-CNN [41] uses support images to help correct the meta-learning method of querying feature vectors.

B. Contrastive Learning

Contrastive learning learns common features of similar instances and distinguishes differences between nonsimilar instances. Unlike generative learning, contrastive learning can not only learn the local features of instances but also distinguish between data at the abstract semantic level of the feature space, so the model and its optimization become simpler and more generalizable. Contrastive learning aims to reduce the distance between positive samples and expand the distance between negative samples, so that the distance between positive samples and anchors is much smaller than between negative samples and anchors, thus achieving the true distance of the original spatial distribution between them. MoCo [17], SimCLR [5], BYOL [28], and MFPC [47] have emerged as the unsupervised representation learning methods that have outperformed supervised learning on some tasks in computer vision. The field of natural language processing (NLP) has recently seen work such as ConSERT [25] and SimCSE [26], using the idea of contrastive learning for sentence representation learning, outperforming SOTA methods on tasks such as semantic text similarity (STS) matching. The Google team [7] has proposed a supervised contrastive learning approach based on this paradigm, further enriching the field of contrastive learning through better pretrained models for downstream tasks. However, all these tasks are often oriented toward classification.

In computer vision, the idea of supervised contrastive learning has been introduced to FSOD [7], [19], while in the field of remote sensing, to our knowledge, this has not been proposed.

III. Method

We designed a two-stage training process, for FSOD [48], [49], [50]. Before we start training, the network automatically learns a more general data augmentation strategy to apply to the feature distribution of the dataset under different domains. We provide an efficient set of intelligent data augmentation combinations for training and downstream tasks, introduce a contrastive learning RoI head, and use the proposal feature in object detection as a positive and negative pair of samples to jointly optimize the CPE loss [6], breaking the bottleneck of FSOD in remote sensing. A general overview of our approach is shown in Fig. 2.

A. Preface

The FSOD algorithm is mature and can learn transferable knowledge from natural images with few samples. However, designing object detection algorithms for remote sensing datasets is challenging due to object size, scene complexity, and annotation costs. A robust and well-generalized FSOD algorithm is needed. The existing FSOD algorithms do not focus on data augmentation, which can be simple and effective without adding computational burden to the network. We introduce an autonomous learning algorithm, autonomous augmentation-aware search (AAAS), to search the optimal combination of data augmentation strategies [9]. The algorithm yields the optimal set of enhancement strategies to provide a more intelligent integration of data enhancement for detection tasks in remote sensing for FSOD. In addition, complex scenarios, too small sample sizes, and large differences in target sizes can lead to accurate localization but wrong classifications in the novel class test, which is the main reason for the unsatisfactory results of the common FSOD baseline application on the remote sensing datasets. We bring invisible similar features closer together and push dissimilar ones farther apart by adding a contrastive classification head to the proposal features of two-stage regression [6]. We refer to this method of using a contrastive learning classification head for feature clustering (CLFC). We combine these methods, as shown in Fig. 2.

B. Autonomous Augmentation-Aware Strategy

The autonomous augmented search algorithms involve three main components: the search space, search algorithm, and evaluation metric [10], [14], [15], [18], [22], [33], [34], [35]. We design the search space to contain a combination of image-level and box-level augmentation strategies, apply them to remote sensing, and introduce a means to perceive objects of different sizes [9]. The framework of the search algorithm is then analyzed, and we present metrics to evaluate the augmentation strategy.

1) Image-Level Augmentation: The FPN is widely used. To cope with diverse image sizes in the remote sensing datasets and match the structure of common FPN models in both the phases, we introduce searchable image-level zoom-in and zoom-out augmentation, whose operations rely on probability Prob and scale ratio r. Prob_{out} is the probability of taking a zoom-out operation, Prob_{in} is the probability of taking a zoom-in operation, and Prob_{ori} is the probability of taking no operation, where Prob_{out} and Prob_{in} range from 0 to 0.5, and

$$\text{Prob}_{\text{ori}} = 1 - \text{Prob}_{\text{in}} - \text{Prob}_{\text{out}}. \quad (1)$$

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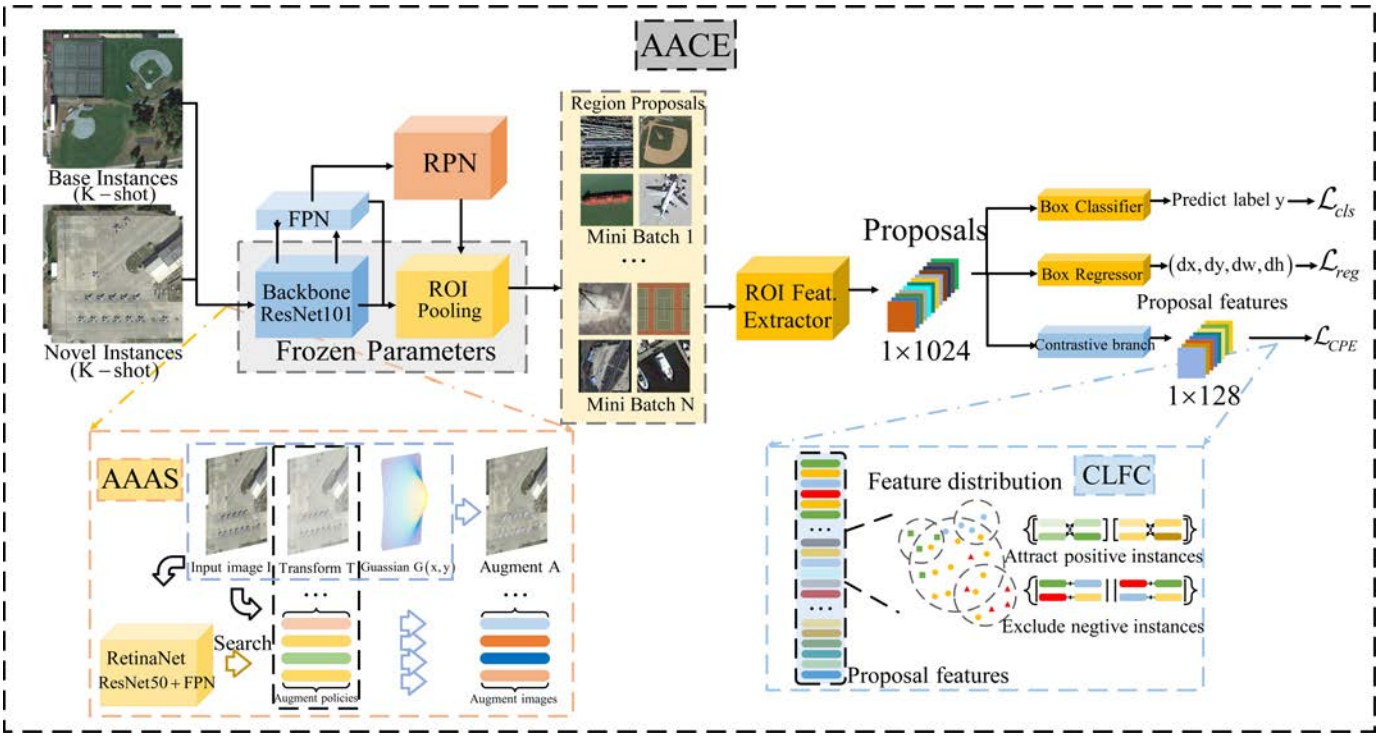


Fig. 2. AACE framework. We use AAAS to search for the optimal set of data augmentation policies, based on which we show the effect of the brightness augmentation approach and introduce it in the main framework of FSOD. Contrastive branch, using the CLFC module, introducing CPE loss to optimize the model together with classification and regression loss.

The scale S indicates the degree of magnification or reduction, i.e., how large or small it is in relation to the original image. During zoom-in, $\{r_{\text{in}}$ is set between 1.0 and 1.5, and for zoom-out, r_{out} is set from 0.5 to 1.0. The new image size is obtained by multiplying the original image size by $\{r_{\text{in}}, r_{\text{out}}\}$.

2) Box-Level Augmentation: In the remote sensing object detection datasets, where the object often has a complex background, simple image-level data enhancement is often insufficient for the detection task. Therefore, we focus on box-level augmentation. Previous approaches to data augmentation [37] have enhanced the object box region in the annotation information. However, this results in a large boundary gap between the augmented and original regions. We use Gaussian distribution augmentation, which smooths out the edge boundaries of the object box area in a natural manner [9]. We choose a brightly colored box-level augmentation strategy, as shown in Fig. 2, and the enhanced image pixel points are

$$I_{\text{gauss}} = G(x, y) \cdot I_{\text{ori}} + (1 - G(x, y)) \cdot I_{\text{rec}} \tag{2}$$

where I_{gauss} refers to the image after Gaussian box-level augmentation, I_{ori} is the original image, I_{rec} is an image obtained by box-level augmentation of the previous rectangular box, and $G(x, y)$ is the Gaussian distribution. For an object bounding box of known size $H \times W$, its information is (x_c, y_c, h, w), where (x_c, y_c) is the center point coordinate, (h, w) refers to the height and width, respectively, of the object box, and

$$G(x, y) = \exp\left(-\left(\frac{(x - x_c)^2}{2\sigma_x^2} + \frac{(y - y_c)^2}{2\sigma_y^2}\right)\right)$$
$$\sigma_x^2 = h^2 \cdot \frac{W}{2\pi H}, \sigma_x^2 = w^2 \cdot \frac{H}{2\pi W}. \tag{3}$$

We introduce the scale-aware concept [9] through the parameter θ in the Gaussian distribution, which is the ratio of the area occupied by the Gaussian distribution S_{gauss} to that of the original object box S_{rec}; we summarize this and update the variance in (3)

$$\theta = \frac{S_{\text{gauss}}}{S_{\text{rec}}} = \frac{\int_0^H \int_0^W G(x, y)dxdy}{(h \cdot w)}$$
$$\sigma_x^2 = h^2 \cdot \frac{W}{2\pi H} \cdot \theta, \sigma_y^2 = w^2 \cdot \frac{H}{2\pi W} \cdot \theta. \tag{4}$$

We applied scale-aware to the remote sensing domain FSOD, performed experiments on it, and found that it helps base class data, but for smaller amounts of novel class data, this strategy has a negative effect. Our main data augmentation approach therefore consists of image-level and box-level augmentation. For image-level augmentation, we use both zoom-in and zoom-out operations. For box-level augmentation, we use five subpolicies, each containing color and geometric operations, with the corresponding probability and value ranges [9], [37]. In addition, each subpolicy has a corresponding three-scale ruler stock perception parameter for correction. We used a search strategy to search for augmentation policies suitable for the remote sensing dataset in Table I.

3) Search Algorithm: We need to allow the network to automatically learn the parameters that fit the set of policies for the remote sensing dataset. We adopt the tournament selection algorithm [3] for the core of our search [42]. We randomly select $|P|$ policy collections to participate in the evolution from a search space composed of random combinations of the above augmented policies at each iteration. The previous top-k policy is selected as the parent for the next generation of updates. The

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TABLE I
SEARCHED OPTIMAL AUGMENTATION POLICIES

Image-level	(Zoom-in, 0.40, 3)	(Zoom-out, 0.45, 1)
Box-level	Color operations	Geometric operations
Sub-policy1	(Sharpness, 0.7, 6)	(ShearY, 0.8, 2)
Sub-policy2	(Equalize, 0.9, 9)	(ShearX, 0.2, 4)
Sub-policy3	(AutoContrast, 0.1, 9)	(TranslateX, 0.9,6)
Sub-policy4	(Contrast, 0.5, 6)	(ShearY, 0.6, 0)
Sub-policy5	(Equalize, 0.7, 4)	(ShearY, 0.9, 8)
Area-ratio	Small - 0.6 Middle - 10	Large - 0.4

parent policies are cross-mutated to generate new children, and this continues for a specified number of evolutions. In the evolution process, we need to train a base model without data augmentation and fine-tune our model by introducing our evolved set of augmented policies and recording the changes in policies and losses for each generation. The optimal set of augmented policies for the remote sensing dataset is obtained by comparing the test mean average precision (mAP) of different policies using predetermined rewards and penalties for screening turnover [9].

C. Optimized Two-Stage Fine-Tune Method

TFA [4] is a powerful and sustainable method in few-shot learning. TFA starts with two-stage pretraining using base class data and freezes the rest of the network layers, keeping only the last two linear layers to fine-tune our model. Because the frozen network parameters have only been exposed to base class data, the method can easily overfit the base class data during fine-tune. Considering that the FPN structure only contains semantic information at multiple scales of the base class data, and the RPN and RoI feature extractors have been exposed to object frame information related to the base class, the results are often poor when migrating to the detection task of novel class samples, especially for RSIs with more complex scene information. Therefore, we need to consider how the models in two-stage few-shot are migrated to the novel class of detection.

1) Rethinking the Fine-Tuning Approach: The performance of few-shot detection is often determined by the localization and classification effectiveness of the region proposals generated by the RPN, but to fine-tune with few novel class samples presents a greater difficulty in the design of RPN and RoI than pretraining with a large number of base class samples. The number of proposals for positive samples in the novel class decreases significantly during fine-tune, and the scores of anchors for novel targets are generally very low after it starts [6]. This leads to a large number of positive novel samples being filtered out by nonmax suppression (NMS), resulting in ineffective learning of RoI for novel objects. We need to compensate for these novel proposals that are important but have low scores and give more novel proposals the opportunity to be learned by RoI. We also need to balance the number of novel proposals to ensure a reasonable recall rate and effectively improve the classification performance of RoI.

2) Two Optimization Strategies: To address the above issues, we adjust the NMS parameter in the fine-tune phase to retain twice the number of proposals, which ensures that more foreground proposals containing novel objects are retained. The number of samples sent to the RoI proposal is then halved, and these proposals contain only background. From Table VII, we find that this approach breaks through the bottleneck of the traditional TFA applications in remote sensing and sets the stage for the introduction of contrastive learning, providing more valid samples for clustering.

D. Contrastive Learning

In the common two-stage object detection process, autonomously augmented features are used as input to the RPN to generate region proposals, and RoI takes these as input, classifies region proposals, and regresses the object box information. The classical Faster R-CNN pools these proposals in a fixed-size layer and encodes them as RoI features. However, since a small number of samples cannot be trained to learn a more robust representation of novel class features in a few-shot task, a classification method is needed that can distinguish features between categories to achieve better classification accuracy. The traditional class-level margin classification head [30], [31] is clearly inadequate for this, while contrastive learning [7] does not need to focus on the tedious details on instances, but only on the feature space at the abstract semantic level, which is good for complex scene information and the few samples in RSIs. We add a contrastive learning branch combined with an RoI classification head to jointly perform classification and regression.

1) Contrastive Learning Branch: Considering that the RoI feature extractor contains a ReLU layer, and therefore some of the pixel points in the feature will be zeroed out, resulting in some information loss, we need to use multilayer perceptrons (MLPs) to form a contrastive learning branch. This allows us to encode the RoI features for conversion to contrastive learning features. We then calculate the similarity scores of the different object proposal embeddings [19], [38] and optimize the training process using the contrastive loss function. As the training process progresses, RoI brings the proposal embeddings closer together for the same class and takes them farther apart for different classes. In the fine-tune phase, data from each class with few samples are more closely clustered in the feature space and more clearly delineated from the other classes, allowing a model trained only on a large number of base classes to migrate smoothly to the detection of novel classes. We use t-SNE to visualize the distribution of features during the fine-tune process for each class, as shown in Fig. 3.

2) Optimization of the Loss Function: We used the Faster R-CNN two-stage loss function design in the RPN through binary cross-entropy to bifurcate the foreground and target. RoI uses cross-entropy and smooth-$L1$ for classification and regression. In the fine-tune process for the novel class, we reprogram the loss function to introduce CPE loss [6], so as to jointly optimize the fine-tune phase

$$L = L_{\text{cls}}^{\text{rpn}} + L_{\text{cls}}^{\text{roi}} + L_{\text{reg}}^{\text{roi}} + \lambda \cdot L_{\text{CPE}}^{\text{roi}} \tag{5}$$

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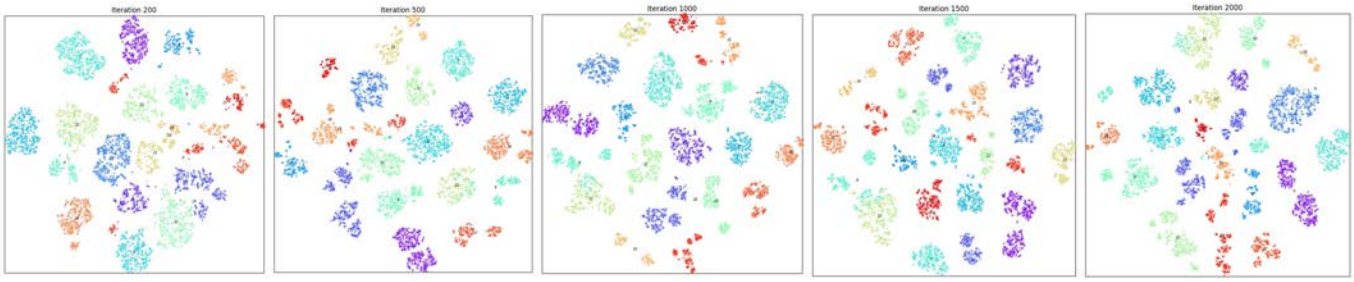


Fig. 3. (Left to right) Results of feature t-SNE visualization for the DIOR dataset at iterations 200, 500, 1000, 1500, and 2000. With training of CPE loss, the distance between same class feature is obviously pulled in, and the interval between classes increases, achieving effect of clustering different classes of proposal features.

where $L_{\text{cls}}^{\text{rpn}}$ is the binary cross-entropy loss used in the RPN; $L_{\text{cls}}^{\text{roi}}$ and $L_{\text{reg}}^{\text{roi}}$ are the cross-entropy and smooth-$L1$ loss, respectively, used in the RoI; and $L_{\text{CPE}}^{\text{roi}}$ is the CPE loss function introduced in the RoI. The CPE loss requires K RoI object box features f_i in each set of batches, the corresponding intersection-over-union (IOU) score s_i, and the ground-truth category c_i corresponding to each feature

$$L_{\text{CPE}}^{\text{roi}} = \frac{1}{K}\sum_{i=1}^{K}\delta(s_i \geq \varepsilon)\cdot L_{f_i}. \tag{6}$$

To avoid a low-IoU proposal embedding interfering with the clustering effect, we set the constraint function $\delta(x)$ to take the value 1 if the condition is met, and 0 otherwise, and set the threshold ε to 0.7. The more representative L_{f_i} is filtered out to participate in the optimization process

$$L_{f_i} = -\frac{1}{K_{\{s_i=s_j\}}-1}\sum_{j=1,j\neq i}^{K}\delta(s_i=s_j) \cdot \log\frac{\exp(\boldsymbol{f_i^T}\cdot\boldsymbol{f_j}/\tau)}{\sum_{z=1,z\neq i}^{K}\exp(\boldsymbol{f_i^T}\cdot\boldsymbol{f_z}/\tau)} \tag{7}$$

where $K_{s_i=s_j}$ is the number of object box features in a batch with the same label as the current one, and τ is the temperature coefficient in InfoNCE [20]. Together, the above equations form our new loss function, which was found experimentally to enable a more reasonable optimization of the network with CPE loss when λ is 0.5.

IV. Experiments

We conducted extensive experiments to verify our proposed scheme. First, we searched for the optimal set of augmentation policies using DIOR, which is a large dataset. Augmentation policies were transformed into augmentation strategies suitable for remote sensing after cross-mutation and filtering in each iteration. Based on our search results, we added these policy collections for enhanced perception to the standard fine-tune few-shot network and introduced the idea of contrastive learning, which subtly achieves a large margin of few-shot tasks in remote sensing. We conducted ablation experiments on modules of interest and proposed a variety of augmentation strategies that demonstrate the effectiveness of this work.

A. Benchmark Datasets

NWPU VHR 10 [39] is a small-scale, ten-category geospatial object detection dataset. The categories are airplane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, and vehicle. The data consist of 800 VHR optical RSIs, of which 715 color images at 0.5–2-m resolution are from Google Earth, and 85 sharpened color infrared images at 0.08-m resolution are from Vaihingen. The set contains no targets. We divided the data into training and test sets at an 8:2 ratio. To meet the requirements of the few-shot task [21], we considered the classes of airplane, baseball diamond, and tennis court as the novel class, and the rest as the base class. In the fine-tune phase, we randomly sampled the k-shot with parameters $k = 3$, 5, and 10 for the few-shot test.

DIOR [40] is a large-scale benchmark dataset containing 23 463 images and 192 472 instances covering 20 object classes: airplane, airport, baseball field, basketball court, bridge, chimney, dam, expressway service area, expressway toll station, harbor, golf course, ground track field, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and wind mill. The RGB image size is 800 × 800. We used approximately 50% for training and 50% for testing. We used the five classes of airplane, baseball field, tennis court, train station, and wind mill as the novel class, and the remaining 15 classes as the base class [21]. Again, we used $k = 3$, 5, and 10 for the fine-tune phase of testing.

B. Implementation

1) Search for Optimal Augmentation Policies: We selected RetinaNet [2] for the search on ResNet50-FPN [1], [9], using the DIOR dataset, which contains more remote sensing information, for training and testing. Each submodel was fine-tuned with 2000 iterations under a pretrained model, and the search process lasted for ten iterations in the subsequent evolutionary search. The evolutionary population size was set to 50, and the top-ten submodels were taken as the parents of the next evolving generation. After all the evolutionary processes were completed, we chose the optimal data augmentation policies to perform fine-tune in RetinaNet. A multiscale training baseline was adopted, ranging from 640 to 800. Training took place on four V100 GPUs, so the number of images in each batch was set to 8.

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TABLE II

COMPARISON OF FEW-SHOT DETECTION RESULTS FOR NOVEL CLASSES ON THE NWPU VHR 10 DATASET. RESULTS ARE AP50 (%). (∗) REPRESENTS EXPERIMENTAL RESULTS CITING OTHER ARTICLES. (†) DENOTES REPLICATION EXPERIMENT USING DATASET DIVIDED IN THIS ARTICLE

Methods/shot	*RepMet [13]			†Meta R-CNN [36]			†TFA [4]			*FSODM [32]			*FSRW [11]			†FSCE [6]			Ours		
	3	5	10	3	5	10	3	5	10	3	5	10	3	5	10	3	5	10	3	5	10
airplane	19.00	20.00	22.00	23.96	30.82	37.63	20.71	43.44	62.16	15.00	57.00	-	13.00	24.00	20.00	33.51	31.84	37.23	40.85	46.94	49.24
baseball diagmond	36.00	36.00	39.00	57.76	63.28	66.28	57.42	52.37	60.41	57.00	84.00	-	12.00	39.00	74.00	54.07	76.04	78.31	66.77	72.94	71.80
tennis court	12.00	14.00	18.00	37.47	45.72	56.38	27.80	28.67	31.12	25.00	16.00	-	11.00	11.00	26.00	25.33	31.23	39.44	29.66	37.50	42.22
mean	22.33	23.33	26.33	39.73	50.16	53.43	35.31	41.49	51.23	32.33	52.33	-	12.00	24.67	40.00	37.64	46.37	51.66	**45.76**	**52.46**	**54.42**

TABLE III

COMPARISON OF FEW-SHOT DETECTION RESULTS FOR NOVEL CLASSES ON THE DIOR DATASET. RESULTS ARE AP50 (%). (∗) REPRESENTS EXPERIMENTAL RESULTS CITING OTHER ARTICLES. (†) DENOTES A REPLICATION EXPERIMENT USING DATASET DIVIDED IN THIS ARTICLE

Methods/shot	*RepMet [13]			†Meta R-CNN [36]			†TFA [4]			*FSODM [32]			*FSRW [11]			†FSCE [6]			Ours		
	3	5	10	3	5	10	3	5	10	3	5	10	3	5	10	3	5	10	3	5	10
airplane	-	9.00	13.00	17.54	15.37	20.79	3.04	9.63	15.29	-	9.00	16.00	9.00	15.00	19.00	17.56	26.27	32.45	20.05	31.57	35.43
baseball field	-	19.00	33.00	57.42	59.29	61.32	1.96	3.43	5.93	-	27.00	46.00	33.00	45.00	52.00	39.65	44.08	50.28	43.77	47.20	53.21
tennis court	-	11.00	24.00	38.96	52.97	63.22	3.52	15.87	22.61	-	57.00	60.00	47.00	54.00	55.00	53.33	53.15	56.79	56.88	56.90	58.18
train station	-	1.00	1.00	3.10	4.65	6.43	0.01	0.18	2.48	-	11.00	14.00	9.00	7.00	18.00	6.97	11.65	11.82	6.09	10.43	16.74
wind mill	-	1.00	1.00	5.78	4.32	7.29	0.99	1.51	5.24	-	19.00	24.00	13.00	18.00	26.00	5.16	5.67	7.36	6.71	7.75	9.04
mean	-	8.20	14.40	24.56	27.32	31.81	1.90	6.12	10.31	-	24.60	32.00	22.20	28.00	34.00	24.53	28.16	31.74	**26.70**	**30.77**	**34.52**

2) Two-Stage Few-Shot Detection Method: We obtained the optimal augmentation policies in Table I after a search, where (operation, probability, and magnitude) were indicated in the image level and box level, and the area ratio indicates the degree of influence on the size of the scale parameter θ [9]. We applied our searched data augmentation policies to enrich the RSI information based on Faster R-CNN. Here, we chose the most commonly used network structure of ResNet101-FPN. We used four V100 GPUs and selected a batch of 8. To better exploit the potential of different size datasets, for the NWPU VHR 10 dataset, we adjusted the learning rate to 0.002 and iterations to 2000 in the training process of the base class. In the fine-tune process of the novel class, the learning rate was set to 0.002, and iterations were set to 4000. For the DIOR dataset, the learning rate was 0.02, and the number of iterations was 18 000 during base class training. In the fine-tune phase, the learning rate was also 0.02, and the number of iterations was 25 000. All these experiments used a standard SGD with a weight decay of $1e{-}4$ and momentum of 0.9.

C. Comparison With Other Algorithms

1) NWPU VHR 10 Results: The results of this article are shown with the results of other methods in Table II, where the classification accuracy can be specified for each class of the novel class data. We selected the optimal performance of this article for NWPU VHR 10 after sifting through the results of different augmentation strategies and modules. Although some of these methods have better results for one or more of these classes, our method is still superior in terms of overall accuracy. We emphasize that some of these algorithms were derived from our surfacing, specifically using our off-the-shelf partitioned dataset, while others were derived from the results in the original papers, so data partitioning is different to some extent, but this article shows good results on randomly sampled few-shot training samples. This is because airplane is different from other classes, and the network can easily learn the representation of such samples as the amount of training increases. For baseball diamond, this class has a large target and a small feature similarity to some classes in the dataset, so when the number of training samples is ten, the accuracy decreases compared with five. This is because too many training samples will cause overfitting. The scene information around the tennis court class is simpler, and the feature details of the target are more consistent, so it is easy for the network to learn stable and robust information. Therefore, for the same large objects, as the number of shots increases, so does the accuracy.

2) DIOR Results: The DIOR dataset is larger, and it contains richer information, so the accuracy of each novel class increases with the number of shots, because the model is more stable when trained from a large dataset, and can avoid interference caused by scene information. The accuracy of train station and windmill is lower for different numbers of shots because of the greater distance in the feature space from the base class data in Table III. Therefore, if the accuracy of these two classes can be solved in the future, the accuracy of the novel class will definitely be improved significantly.

D. Ablation

1) Analysis of Augmentation Strategies: To make more rational use of the augmentation-aware policy, we designed three augmentation strategies for the few-shot novel class detection. Extensive experiments are used to demonstrate the effectiveness of the augmentation policies applied at different stages. There are two phases, training for a large amount of base class data and fine-tune for a small amount of novel

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TABLE IV

AUGMENTATION STRATEGIES FOR FEW-SHOT NOVEL CLASS DETECTION. RESULTS OF NWPU VHR 10 AND DIOR ARE MAP50 (%)

Strategies	NWPU VHR 10			DIOR		
	3-shot	5-shot	10-shot	3-shot	5-shot	10-shot
Baseline	37.64	46.37	51.66	24.53	28.16	31.74
with AT	35.37	45.01	53.58	23.33	28.94	33.98
with AF	43.37	**50.76**	**53.75**	25.91	**30.00**	**34.43**
with AT+AF	**45.40**	46.36	51.42	**26.27**	29.51	34.35

TABLE V

AUGMENTATION METHODS FOR FEW-SHOT BASE CLASS DETECTION WITH AT. RESULTS OF NWPU VHR 10 AND DIOR ARE MAP50 (%), WHERE (+) MEANS CUMULATIVE

Methods	NWPU VHR 10	DIOR
Baseline	89.08	72.33
+Image-level	**92.78**	74.00
+Box-level	92.51	74.13
+Scale param θ	92.20	**74.24**

class data, and we applied the augmentation policy to these two phases separately, referred to as augmentation for base training (AT) and augmentation for novel fine-tune (AF), respectively, designed experiments to apply the policy to both the phases, and compared the results with the baseline. The final results are shown in Table IV, from which it can be seen that the optimal choice of our proposed augmentation policy is different for the two datasets. It can be seen that the AT strategy is not entirely positive, with the smaller NWPU VHR 10 having a disruptive effect on AT at three- and five-shot, and the network needs to receive more information about the novel class to make the model converge better on the novel class. For the larger DIOR, only three-shot is still negatively affected, suggesting that the larger dataset better fits the AT strategy, because the AF strategy is applied directly to the fine-tune stage, a strong data augmentation that makes it easier for the network to focus on the key regions of the test images. In contrast, when we use the AT and AF strategies together, the performance is similar in the two datasets, but there are variations in different shots. To pursue optimal accuracy, a combination of AT and AF strategies can be used for the three-shot of both the datasets, while for five- and ten-shot, the AF strategy is chosen.

2) Augmentation Methods: The augmentation policy in this article has three modalities: image-level, target-level, and a scale-aware parameter [9]. For different datasets, we conducted ablation experiments for the base and novel classes with the AT and AF augmentation strategies, respectively, with the results as shown in Tables V and VI. In Table VI, for the NWPU VHR 10 dataset, three-shot is more suitable for image-level augmentation, while for five- and ten-shot, we should take a combination of image- and box-level augmentation as the amount of novel class training data increases. For the DIOR dataset, three- and five-shot are enhanced at the image level, while for ten-shot, we chose a combination of image and box. We believe that for a small number of images, too much attention to the detailed box information will affect the training

TABLE VI

AUGMENTATION METHODS FOR FEW-SHOT NOVEL CLASS DETECTION WITH AF. RESULTS OF NWPU VHR 10 AND DIOR ARE MAP50 (%), WHERE (+) MEANS CUMULATIVE

Methods	NWPU VHR 10			DIOR		
	3-shot	5-shot	10-shot	3-shot	5-shot	10-shot
Baseline	37.64	46.37	51.66	24.53	28.16	31.74
+Image-level	**45.76**	48.57	51.45	**26.70**	**30.77**	34.13
+Box-level	45.65	**52.46**	**54.42**	25.94	30.36	**34.52**
+Scale param θ	43.37	50.76	53.75	25.91	30.00	34.43

effect of the network, while as the number of images increases, the detailed information of the object box will help the network learn more powerful features. The introduction of the scale parameter θ is not applicable to the detection of the novel class for few-shot RSIs, but will play a negative role, which may be related to factors such as complex background information of RSIs and incongruent sizes. In contrast, in Table V, we find that the optimal strategy for NWPU VHR 10 still takes the image level, while DIOR requires the introduction of the scale parameter θ. We believe that for large-scale datasets that contain more information, the conventional AT strategy requires the scale parameter θ to learn more detailed features.

3) Impact of Different Modules: We used the fine-tune means of the few-shot task as a benchmark [4], and first revised the RPN and RoI in sequence. It can be seen that all the groups of experiments in Table VII show a significant effect of the modules used in AACE, except for three-shot in the smaller-scale NWPU VHR 10, where the effect of the corrected RoI fluctuates slightly, especially for TFA on DIOR, with a maximum 14.53% improvement. Furthermore, the CPE loss [6] was introduced, and the function was found to be a good fit for the contrastive proposal encoding used in this article, achieving an optimal 8.94% improvement on the DIOR dataset. Finally, with the introduction of our strong aug-aware augmentation strategy, the experimental results significantly surpassed the TFA benchmark, further validating that this article effectively combines the various modules and successfully applies them to the task of FSOD in RSIs.

4) ImageNet Pretrained Weights: ImageNet [43], the definitive dataset benchmark for natural images, relies on its pretrained weights for much work. In the field of remote sensing, it is difficult to find a dataset benchmark of this scale that can replace it. We address the few-shot task in remote sensing, so we introduced a comparative experiment to verify that our article can be better applied to remote sensing than loading ImageNet pretrained weights. We loaded ImageNet weights directly in the fine-tune phase for comparison with our article, and we present the results for the novel class in Table VIII, from which it can be seen that for the same experimental parameters, our work converges faster and with better accuracy on both the datasets than simply loading ImageNet weights.

E. Analysis of Visualization Results

1) T-SNE Visualization: In Fig. 3, we selected more data-rich DIOR for visualization experiments and

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TABLE VII

ROLES PLAYED BY MODULES IN AACE (OURS) FOR FEW-SHOT NOVEL CLASS OBJECT DETECTION. RESULTS OF NWPU VHR 10 AND DIOR ARE MAP50 (%), WHERE (+) MEANS CUMULATIVE

Methods	Revision		CPE	Augmentation	NWPU VHR 10			DIOR		
	RPN	RoI	Loss	Strategy	3-shot	5-shot	10-shot	3-shot	5-shot	10-shot
TFA[4]	✗	✗	✗	✗	35.31	41.49	51.23	1.90	6.12	10.31
AACE (Ours)	✔	✗	✗	✗	36.81	40.85	50.71	6.49	12.53	17.36
	✔	✔	✗	✗	36.70	42.02	51.18	15.59	20.65	24.82
	✔	✔	✔	✗	37.64	46.37	51.66	24.53	28.16	31.74
	✔	✔	✔	✔	**45.76**	**52.06**	**54.42**	**26.70**	**30.77**	**34.52**



Fig. 4. Detection results of the proposed FSOD framework (AACE). (Top) Base class. (Bottom) Novel class.

TABLE VIII

PRETRAINED WEIGHTS FOR FEW-SHOT NOVEL CLASS DETECTION. RESULTS OF NWPU VHR 10 AND DIOR ARE MAP50 (%)

Pretrained on	NWPU VHR 10			DIOR		
	3-shot	5-shot	10-shot	3-shot	5-shot	10-shot
ImageNet	27.76	29.67	43.67	6.86	9.8	13.7
Our work	**45.76**	**52.06**	**54.42**	**26.70**	**30.77**	**34.52**

TABLE IX

PERFORMANCE METRICS OF ONE-STAGE AND TWO-STAGE FSOD DETECTION METHODS

Performance Metric	One-stage [11]	Two-stage(Our work)
Flops	71.47G	31.64G
Params quantity	66.29M	0.59M
Forward time	64.46ms	220.84ms

demonstrated that CPE loss can better cluster similarity fuzzy feature pairs. This clustering involves three steps: designing a way to extract proposal features in the fine-tuning stage, decreasing the distance between positive samples and increasing the distance between negative samples, and solving the problem that while the novel class targets are located accurately, their classification is interfered with by the base class in the few-shot task in remote sensing.

2) Visualization of AACE Results: To better demonstrate the effectiveness of our proposed AACE on remote sensing domain datasets, we selected some test plots for the base and novel classes for visualization of object boxes and confidence levels, as shown in Fig. 4. It can be seen from the base class data that after a large amount of data training, good localization and classification performance can be achieved in the few-shot fine-tune stage. However, in the first figure of the base-class results, due to the complex background information of the RSI, only part of the vehicle information can be detected, and the large targets stadium and ground track field are not affected by this. For the novel class with a small number of samples, AACE can still achieve good classification performance, and most of the samples can be successfully localized and classified accurately. This demonstrates that this article offers a reasonable solution to the few-shot problem in the remote sensing field, and that the model trained on a large amount of base data can be successfully transferred to the detection of a small number of novel samples.

F. Performance Analysis

To help readers understand the performance differences between one-stage and two-stage methods in the field of FSOD and to gain insight into the performance details of our work, we conducted additional comparative experiments. Our work adopts the two-stage method, and we selected a one-stage FSOD work [11] to compare with it. In this

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study, we fixed the input to randomly generated data of size $1 \times 3 \times 800 \times 800$ and conducted experiments on a single V100 device. We calculated the parameters, FLOPs, and forward propagation time, and summarized the results in Table IX. The table shows that compared with the one-stage method, although our work has longer forward propagation time, it has fewer parameters, lower FLOPs, and less computational burden.

V. Conclusion

We introduced a contrastive learning paradigm based on data augmentation perception, AACE, and applied it to the few-shot task in the field of remote sensing. Faced with complex scene information and few training samples, we conducted a search for the optimal augmentation policy on the dataset and cross-mutation through continuous iterations. We introduced the idea of contrastive learning by designing a reasonable RoI to bring the positive samples closer and push negative samples further away, to better cluster the proposal features with fuzzy decision boundaries, and combined intelligent data augmentation with contrastive proposal embedding in two remote sensing datasets, NWPU VHR 10 and DIOR, whose final combination achieved good results on the NWPU VHR 10 and DIOR datasets.

Based on our experiments, we observed that certain classes had low precision, which consequently impacted the overall precision of the novel class. To address this, we plan to conduct a thorough analysis of these specific categories, exploring their feature distribution and potential semantic information. By doing so, we anticipate enhancing the accuracy of FSOD for the novel class.

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